

Spectrum-Matched Quantum-Assisted PINNs: A Constructive Design Methodology for Problem-Specific Variational Circuits, Instrumented by Explainable-AI Diagnostics

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Abstract

Physics-informed neural networks (PINNs) learn low frequencies fast and high frequencies slowly (spectral bias), because their neural tangent kernel (NTK) spectrum decays with frequency. A data-re-uploading quantum circuit behaves differently: its output is a Fourier series whose frequency set is fixed, exactly, by its encoding. We join the two facts into a constructive design rule, Spectrum-Matched Circuit Design (SMCD): read the frequencies a PDE’s solution needs from the operator’s Fourier symbol and the forcing, then build the circuit whose frequency set covers them, with no architecture search. The hybrid model mixes K copies of the designed circuit through a linear head. Six explainable-AI instruments, run identically on matched classical and hybrid models, test the mechanism rather than only the accuracy, and every claim is pre-registered with a numeric threshold and scored by a script. The circuit produces exactly its designed frequencies (leakage below 4×10^{-15}). Designed frequencies beat random ones of the same size on the heat equation ($6.5\times$, one copy), on Poisson ($236\times$, eight copies) and on a groundwater problem ($9.9\times$, eight copies), on every seed. On waterlogging of farmland from leaking canals, the eight-copy circuit reaches 0.09% error, $18\times$ more accurate than a classical network of the same size, and places the waterlogged stretch to within 1 m. The instruments explain the mechanism: 99% of the improvement lies inside the designed frequency set, where the NTK is flatter than outside, and copies matter because in a single circuit every Fourier amplitude depends on the same angles, so the optimiser trades the slow wave for the fast one; eight copies cut the Poisson error by almost three orders of magnitude.

1 Introduction

A classical physics-informed neural network (PINN) learns low-frequency solution components fast and high-frequency components slowly — spectral bias — because its neural tangent kernel (NTK) eigen-spectrum decays with Fourier-mode index. Separately, a data-re-uploading variational quantum circuit realises a truncated Fourier series whose frequency set is determined exactly by the eigenvalues of its data-encoding Hamiltonians [10]. Prior hybrid quantum PINN work [2, 6, 11, 14] reports accuracy on hand-picked circuit sizes, without a principled way to choose the circuit *given* the PDE.

This paper joins the two facts. Given a PDE, the target spectral support is computed from the operator’s Fourier symbol and the forcing and boundary data; the circuit’s encoding scalings, re-uploading depth, qubit count and entangler then follow by formula so that its frequency set covers that support. We call this Spectrum-Matched Circuit Design (SMCD, Section 3). The hybrid model mixes K copies of the designed circuit through a linear head: every copy has the designed frequencies, and the head sets their amplitudes independently.

A design rule is only useful if it can be tested, so the project is built around measurement. Six explainable-AI instruments run at the same checkpoints on matched classical and hybrid models (Section 4), so any difference can be traced to the quantum layer. Every claim about the method was written down with a numeric threshold and committed to git before the runs that test it, and a script issues every verdict.

Contributions.

1. **SMCD, a constructive map from PDE to circuit** (Section 3): four propositions and an algorithm that output the encoding scalings, depth, qubit count and entangler deterministically, with a design card that can be checked before training.
2. **A hybrid NTK decomposition** for models with disjoint classical and quantum parameters, $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q$, with no cross term (Section 3.5).
3. **Circuit copies**, a model family in which K copies of the designed circuit share its frequency set but have independent amplitudes (Section 3.8).
4. **An explainable-AI protocol for hybrid PINNs**: six instruments with published metric definitions, run identically on every model family (Section 4).
5. **Evidence** on the Poisson and heat equations and on an applied problem, groundwater under irrigated farmland (Sections 5–6).

Findings in brief. The circuit realises exactly the frequencies SMCD designs, and the design matters: against the same circuit with random frequencies, the designed circuit is $6.5\times$ more accurate on heat, $236\times$ on Poisson and $9.9\times$ on groundwater, on every seed. The instruments show why it works (the improvement lies inside the designed frequency set, where the NTK is flat) and why several copies of the circuit train far better than one. On waterlogging of farmland, eight copies reach 0.09% error, $18\times$ more accurate than a classical network of the same size, and place the waterlogged stretch to within 1 m. SMCD also predicts, before training, when a quantum layer will not help.

2 Background

This section states the results the design rule builds on; full derivations are in the repository’s derivations folder.

2.1 Classical side

PINNs. A PINN [9] minimises the PDE residual at sampled collocation points, with derivatives from automatic differentiation that are exact to floating-point precision. Unlike a Galerkin or finite-element method, the optimisation is non-convex in the network parameters even when the PDE is linear.

The NTK and spectral bias. Writing $\Theta(x, x') = \nabla_{\theta} N[u_{\theta}](x) \cdot \nabla_{\theta} N[u_{\theta}](x')^{\top}$ for the residual’s tangent kernel [5], gradient-flow training in the lazy regime linearises to $\dot{e} = -\Theta_0 e$, so each eigenmode of Θ_0 decays as $\exp(-\lambda_i t)$. For a tanh or ReLU MLP the eigenvalues decay roughly as a power law in Fourier-mode index, so high-frequency components are learned last and slowly (Figure 1). Fourier-feature encodings [12] are the standard classical fix, and therefore a necessary baseline: a quantum layer that beats only a plain MLP has bought nothing Fourier features do not already supply.

Hard boundary conditions. The ansatz $u = B(x) + D(x)N(x)$, with D vanishing on the boundary and B matching the boundary data, satisfies the boundary conditions by construction and removes the boundary-loss block of the NTK.

2.2 Quantum side

Re-uploading circuits are Fourier series. For encoding gates $\exp(-ixH^{(l)})$ with generator eigenvalues $\{\lambda_a^{(l)}\}$, an L -layer re-uploading circuit’s output is a truncated Fourier series whose frequencies are the signed sums of per-layer eigenvalue differences [10]. This is the fact SMCD engineers on (Proposition 1).

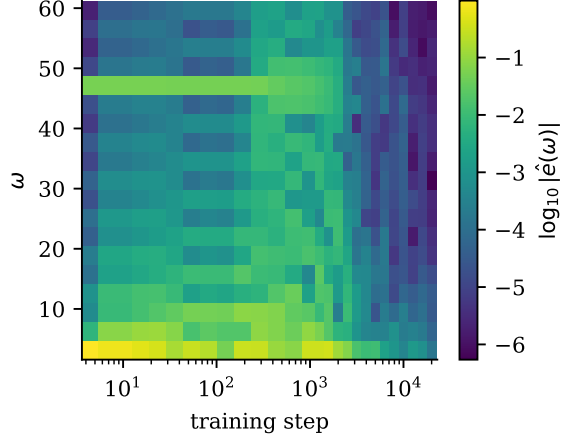


Figure 1: Spectral bias, measured: per-frequency error of a plain MLP over training on the Poisson problem, $u^* = \sin(\pi x) + \alpha \sin(15\pi x)$ with $\alpha = 0.05$. The $\omega = \pi$ mode converges first; $\omega = 15\pi \approx 47$ needs 16.2 times as many steps to reach tolerance (3 seeds).

Exact derivatives on hardware. The parameter-shift rule $\partial_\theta f = \frac{1}{2}[f(\theta + \pi/2) - f(\theta - \pi/2)]$ holds for Pauli-generated gates, and because the input x enters as a rotation angle, $\partial_x f$ is also exact by a shift. The PINN residual can therefore be computed exactly on quantum hardware, not only in simulation.

Barren plateaus. $\text{Var}[\partial_\theta C]$ decays exponentially in qubit count for a global cost on a deep random circuit [7]. SMCD avoids this by construction: the depth rule picks the *minimum* depth that covers the target band, the observable is local, and angles start small.

Effective dimension. The effective dimension of Abbas et al. [1], from the empirical Fisher spectrum, measures capacity beyond parameter count and is reported for every family.

3 Methodology: Spectrum-Matched Circuit Design (SMCD)

3.1 Setup

The hybrid ansatz used throughout (the *serial* form; Section 5 compares it with a random-scaling variant) is

$$x \xrightarrow{E_\varphi} z \in \mathbb{R}^m \xrightarrow{Q_\theta} \phi(z) \in \mathbb{R} \xrightarrow{w} u(x),$$

where E_φ is an affine classical encoder, Q_θ a data-re-uploading variational quantum circuit, and w a linear head. The split is deliberate and is itself a claim: the classical part learns the coordinate *warping*, the quantum part supplies the Fourier *basis*, and the head learns the *coefficients* — exactly the structure of a spectral method with a learned coordinate map.

3.2 Proposition 1 — Circuit spectrum

Proposition 1 (Circuit spectrum). *An L -layer data-re-uploading circuit, with encoding gate $\exp(-ix H^{(l)})$ at layer l for Hermitian generator $H^{(l)} = \sum_a \lambda_a^{(l)} |a\rangle\langle a|$, realizes an observable expectation of the form*

$$f(x) = \sum_{\omega \in \Omega} c_\omega(\theta) e^{i\omega x}, \quad \Omega = \left\{ \sum_{l=1}^L \left(\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)} \right) \right\},$$

i.e. a trigonometric polynomial whose frequency support is the set of all signed differences of encoding-generator eigenvalues, summed over layers.

Proof sketch. Expand $f(x) = \langle 0|U^\dagger(x, \theta) M U(x, \theta)|0\rangle$ by inserting the eigenbasis resolution of each diagonal encoding gate independently on the bra and ket sides. Every (bra-path, ket-path) pair contributes an x -dependent phase $\exp(ix \sum_l (\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)}))$ multiplied by an x -independent amplitude collecting the trainable gates and M along that path; collecting paths by total frequency gives the boxed form. The full derivation, worked first for one qubit/one layer and then generalized to n qubits and d -dimensional inputs, is in docs/derivations/07_circuit_fourier_spectrum.md. ■

Two subtleties that change the constructive guarantee. Two facts, easy to get wrong and verified numerically against both an independent statevector simulator and a PennyLane oracle (agreement $< 10^{-10}$) before being trusted:

1. **Generator-vs-frequency factor of two.** The generator’s eigenvalues are $\pm\omega/2$ for an $RZ(\omega x)$ gate, but the *realized* frequency in $f(x)$ is the full ω , not $\omega/2$: the eigenvalue-difference structure above supplies a compensating factor of two. Reading the generator’s eigenvalues off the gate and reporting that as the frequency halves every number in the design.
2. **State-preparation degeneracy.** If the qubit is prepared exactly at the equator ($RY(\pi/2)$ before the first encoding gate, an equal superposition), every frequency whose balanced-ternary leading digit is zero — exactly one third of Ω — has identically zero amplitude for *every* trainable-parameter setting, independent of ansatz expressivity. This is not a numerical inconvenience: it makes Proposition 3 below false for any target spectrum touching a dead frequency under the literal $\pi/2$ prescription. We use $\phi_{\text{prep}} = \pi/3$ throughout, verified to have no dead frequencies.

3.2.1 Corollary: ternary scaling and the depth rule

Set the per-layer scalings to $\omega_l = \Delta \cdot 3^{l-1}$ (single-wire case). Then $\Omega = \Delta \cdot \{-\frac{3^L-1}{2}, \dots, \frac{3^L-1}{2}\}$ — exactly 3^L distinct frequencies with no degeneracy, a bijection with the balanced-ternary representation of the integers in that range (proof: substitute $m_l = d_l - 1$ to convert the signed sum into an ordinary base-3 expansion shifted by $\frac{3^L-1}{2}$). This grows *exponentially* in depth, versus $2L + 1$ distinct frequencies for linear scaling $\omega_l = \omega_0$ — the design knob almost every prior re-uploading paper leaves fixed.

To guarantee the reachable band covers a target maximum frequency K at ternary spacing Δ :

$$\max |\Omega| \geq K \iff \Delta \frac{3^L-1}{2} \geq K \iff L \geq \log_3 \left(\frac{2K}{\Delta} + 1 \right) \implies \boxed{L = \lceil \log_3 \left(\frac{2K}{\Delta} + 1 \right) \rceil}.$$

Both the factor of two inside the logarithm and the $+1$ matter. For the Poisson target spectrum ($K = 15\pi$, $\Delta = \pi$), the naive $L = \lceil \log_3(K/\Delta) \rceil$ gives $L = 3$, whose reachable band tops out at $13\pi < 15\pi$ and silently misses the target mode; the rule above gives $L = 4$, whose band reaches $40\pi \geq 15\pi$.

3.3 Proposition 2 — PDE target spectrum

Proposition 2 (PDE target spectrum). *For a linear constant-coefficient operator $\mathcal{L}$ with Fourier symbol $\sigma(k)$, the solution of $\mathcal{L}u = f$ on a periodic/rectangular domain satisfies $\hat{u}(k) = \hat{f}(k)/\sigma(k)$. The essential spectral support $S_\varepsilon = \{k : |\hat{u}(k)| > \varepsilon \|\hat{u}\|\}$ is therefore computable before training, from $\hat{f}$ and σ alone.*

For Helmholtz, $\sigma(k) = k^2 - |\mathbf{k}|^2$, so S_ε concentrates near $|\mathbf{k}| \approx k$ — the near-resonant band a classical PINN’s low-frequency-first learning dynamics reach last, if at all. For the heat equation, σ grows quadratically with $|\mathbf{k}|$ and the solution decays exponentially in time at every nonzero mode, so S_ε collapses toward $k = 0$ as t increases: this is the project’s negative control (Section 5.3) — there is little high-frequency content for a quantum layer to supply, and SMCD predicts this *a priori*, not just observes it after the fact.

3.4 Proposition 3 — Matching condition and resource bound

Proposition 3 (Matching & resource bound). *If $S_\varepsilon \subseteq \Omega$, the hybrid model represents the solution to ε accuracy with a linear head — the remaining optimization problem is convex in w . Covering S_ε with maximum frequency K and resolution Δ requires $L \geq \log_3(2K/\Delta + 1)$ layers (ternary scaling, above) and $n \geq \lceil d \rceil$ qubits for d input dimensions, plus ancilla wires for any required cross-frequency terms.*

This turns architecture choice into arithmetic: given (K, Δ, d) read off the PDE’s own symbol and data, (L, n) follow by formula, not by search.

3.5 Proposition 4 — NTK consequence

Proposition 4 (Hybrid NTK decomposition). *For a hybrid model whose parameters $\theta = (\theta_{\text{cl}}, \theta_q)$ partition into disjoint classical and quantum groups,*

$$\Theta_{\text{hyb}}(x, x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x'),$$

*exactly, with **no cross term**; a term of the form $2\Theta_\times$ does not arise for a shared-output, disjoint-parameter model.*

Proof. $\Theta(x, x') = \sum_{k=1}^P \partial_{\theta_k} f(x) \partial_{\theta_k} f(x')$ is a single inner product of gradient *vectors*, summed over the *full* index set of parameters $k = 1, \dots, P$. When that index set is partitioned into two disjoint blocks θ_{cl} and θ_q (a hybrid model’s classical and quantum parameters never overlap), the sum splits additively over the partition:

$$\Theta_{\text{hyb}}(x, x') = \sum_{k \in \text{cl}} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') + \sum_{k \in q} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x').$$

There is no third term: a cross term would only arise from a fundamentally different quantity, such as a covariance between two *separately* defined kernels or the square of a sum of outputs — not from the NTK of a single model with disjoint parameter groups, which this project’s ansatz is. The identity is also checked numerically on random synthetic Jacobians of matched and mismatched sizes; `src/qapinn/xai/ntk.py` implements it. ■

Because Ω (the quantum block’s realized frequency set) is discrete and prescribed by construction, Θ_q has eigen-directions *at* the encoded frequencies with eigenvalues bounded below independently of $|n|$ for $n \in \Omega$ — the exponential/power-law eigenvalue decay in $|k|$ that drives classical spectral bias (Section 2) is replaced by a flat response over Ω . This is the formal statement of “the quantum layer removes spectral bias inside its band,” and it is what Section 5’s NTK spectrum figures measure directly: decay exponent inside Ω versus outside it, for matched classical and hybrid models.

3.6 Algorithm 1 — SMCD

The design card (final step) is what makes the methodology auditable: every run’s predicted coverage can be set against its measured error (Section 5.3).

3.7 Octave-split circuit ensembles

When a single circuit cannot cover $\hat{S}$ within the barren-plateau depth budget ($L > L_{\text{max}}$), SMCD falls back to M shallow circuits, each matched to one octave band $[2^j \Delta, 2^{j+1} \Delta)$ of $\hat{S}$, summed at the head. This keeps every individual circuit shallow (trainable) while the ensemble covers a wide band — multi-resolution analysis with a quantum basis per resolution level.

Algorithm 1 Spectrum-Matched Circuit Design (SMCD)

Require: PDE operator $\mathcal{L}$, forcing f , boundary data g , domain Ω_x , tolerance ε , qubit budget $n_{\max}$

Ensure: encoding generators & scalings $\{\omega\}$, depth L , qubits n , entangler, observable, ansatz form

- 1: **Symbol:** $\sigma(k) \leftarrow$ Fourier symbol of $\mathcal{L}$ (linearize about a base state if nonlinear; Burgers uses the viscous-scale estimate $k_{\max} \sim 1/\sqrt{\nu}$)
 - 2: **Target:** $\hat{S} \leftarrow \{k : |\hat{f}(k)/\sigma(k)| > \varepsilon\} \cup$ boundary-lift spectral support (time-dependent: union over t , record growth rate)
 - 3: **Band:** $K \leftarrow \max |k|$ in $\hat{S}$; $\Delta \leftarrow$ min spacing in $\hat{S}$; $d \leftarrow \dim(x)$
 - 4: **Depth:** $L \leftarrow \lceil \log_3(2K/\Delta + 1) \rceil$ with ternary scalings $\omega_j = \Delta \cdot 3^{j-1}$ (Section 3.2; fall back to linear scaling if $\hat{S}$ is a dense low band)
 - 5: **Width:** $n \leftarrow d + n_{\text{cross}}$, where n_{cross} covers required mixed-frequency terms (e.g. 2-D Helmholtz needs $k_x \pm k_y \Rightarrow$ entangle the two coordinate wires)
 - 6: **if** $n > n_{\max}$ or $L > L_{\max}$ (barren-plateau depth budget) **then**
 - 7: split the band into an octave ensemble (Section 3.7)
 - 8: **end if**
 - 9: **Entangler:** ring-CZ if cross terms are needed, else none (unnecessary entanglement costs gradient variance for no benefit on a separable target)
 - 10: **Observable:** local $\langle Z_0 \rangle$ (never a global Pauli string — barren plateaus)
 - 11: **Ansatz:** hard-constrained BCs, $u = B(x) + D(x) \cdot [w \cdot Q_\theta(E_\varphi(x))]$
 - 12: **Init:** small-angle $\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, $\sigma_\theta = 0.1$; identity-block init for the encoder
 - 13: **Report:** emit a design card ($\hat{S}, \Omega$, coverage $= |\hat{S} \cap \Omega|/|\hat{S}|$, L, n , param count, predicted NTK band), committed alongside every experiment run
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3.8 Circuit copies

A single circuit reads all of its Fourier amplitudes out through one expectation value, so every amplitude is a function of the same few angles: changing one frequency’s amplitude moves the others. The copy model removes that coupling without changing the frequency set. It evaluates K copies of the SMCD circuit on the same encoded input, each with its own trainable angles, and mixes them with a linear head,

$$u(x) = B(x) + D(x) \left[b + \sum_{j=1}^K w_j Q_{\theta_j}(E_\varphi(x)) \right].$$

Every copy has the same encoding scalings, so the model’s frequency set is still exactly Ω and every design-card statement above applies unchanged; only the number of independent handles on the amplitudes grows. $K = 1$ is the plain SMCD circuit. On the one-dimensional problems each copy is a one-qubit, four-layer circuit, and $K = 8$ gives 91 trainable parameters. K and the learning rate were selected on tuning seeds before any held-out run (Section 4.2).

4 Measurement protocol

4.1 Explainable-AI instruments

Six instruments run at the same checkpoints (0, 100, 500, 1k, 5k, 20k steps and the final step) for every model, problem and seed, so a measured difference cannot come from when or how each family was probed (Table 1). Each instrument’s expected signature is stated as a pre-registered claim, which guards against the post-hoc storytelling PINN explainability work is criticised for [8].

Instrument	What it measures	Expected signature for SMCD
NTK spectroscopy [3]	eigenvalues, decay exponent, effective rank and drift $\ \Theta_t - \Theta_0\ _F / \ \Theta_0\ _F$ of the empirical NTK	flatter spectrum inside Ω than outside (Proposition 4)
Per-frequency error	error $u_\theta - u^*$ projected on Fourier modes at every checkpoint	improvement over the MLP concentrated inside Ω
Attribution	Integrated Gradients of the residual over the domain, with completeness violation and correlation to the true residual	attributions track the residual
Fisher information	empirical Fisher spectrum and effective dimension [1]	separates expressivity from parameter count
Gradient variance	$\text{Var}[\partial_\theta L]$ against qubit count and depth	stays above the 2^{-n} barren-plateau line [7]
Probes and landscapes	per-layer ridge-regression R^2 ; 2-D loss slices around the trained point	trained point is a minimum

Table 1: The six instruments, run identically on classical and hybrid models.

4.2 Experimental setup

Problems. Three problems on the unit domain, each with an exact reference: Poisson, $-u'' = f$ with solution $\sin \pi x + 0.3 \sin 15\pi x$, a slow and a fast wave, the case spectral bias makes hard; the heat equation with a 15π ripple in the initial condition, a smoothing operator on which SMCD predicts little benefit; and groundwater flow under irrigated farmland, the application of Section 6.

Model families. Every family uses the same hard boundary ansatz and training loop. The SMCD circuit (one copy, or K copies, Section 3.8); the same circuit with random encoding scalings drawn from $U(0.5, 3)$, the size-matched test of the design itself; an MLP (8,513 parameters) and an MLP of the circuit’s size; a Fourier-feature network (129); and a frequency-matched classical model, a linear head on sine and cosine features at exactly the frequencies SMCD targets, sized to match the circuit. For all three problems SMCD builds a one-qubit, four-layer circuit with $|\Omega| = 81$ frequencies (14–18 parameters per copy).

Training. 20,000 Adam steps followed by 2,000 L-BFGS steps, five seeds per configuration. The one-copy circuit uses seeds 0–4 at a common learning rate. The eight-copy circuit was selected on the Poisson problem with seeds 0–2 from eight configurations ($K \in \{1, 2, 4, 8\}$, two learning rates), while each classical family received eight learning-rate candidates, under a selection rule fixed before any tuning run; all of its reported results are on held-out seeds 10–14, which played no part in selection. Circuits are simulated exactly by a statevector simulator cross-checked against PennyLane to 10^{-10} .

4.3 Pre-registered claims and how they are scored

Every claim about the method was written with a deciding metric and a numeric threshold in the repository’s `docs/` folder and committed to git before the runs that test it; a verification script checks that the committing revision is an ancestor of every such run. Claims about the eight-copy circuit were written after model selection on the tuning seeds and before any held-out run. An adjudication script issues every verdict from the committed run records. A claim is confirmed only when its threshold is met and, for a comparison between two models, all five paired seeds agree (paired Wilcoxon $p \leq 0.0625$, the smallest value five pairs allow). No threshold is changed after a run. The complete set of pre-registered claims and their verdicts is published with the code.

4.4 Implementation

Everything above is implemented in one open codebase (about 5,400 lines of library code, 4,500 of experiment and analysis scripts and 6,300 of tests). Its parts are:

- **A statevector simulator** for data-re-uploading circuits with exact derivatives in the input and the angles, cross-checked against PennyLane to 10^{-10} and against a density-matrix simulation for the noise surrogates.
- **SMCD itself**: the symbol-based target spectrum, the depth and width rules, the octave split and the design card, with a test that every designed circuit emits nothing outside Ω .
- **One training harness** for all seven families (hard or soft boundary conditions, Adam then L-BFGS, deterministic seeding) that records the six instruments at fixed checkpoints and stores every run under a content hash of its configuration.
- **An adjudication pipeline** that reads only the stored run records, issues every verdict, verifies that each claim was committed before its runs, and redraws every figure and table without retraining.

The experiments comprise 327 recorded training runs, about 130 hours of training on one laptop GPU. 402 automated tests and a continuous-integration job rerun the test suite, re-derive every verdict and compare every figure against the committed record on each change.

5 Results

Errors are relative L^2 against the reference solution, reported as the median over five seeds. A relative error of 1 is what the trivial prediction $u \equiv 0$ scores, so the scale runs from 1 (nothing learned) towards 0. The subsections take the evidence one question at a time and say what each result means for the method.

5.1 The circuit builds exactly what SMCD designs

For every one-dimensional problem the SMCD circuit was evaluated at 40 random parameter settings and its output Fourier-transformed. The largest amplitude outside the designed set Ω , relative to the largest overall, is at most 3.8×10^{-15} , which is floating-point zero, and the test suite checks the same property on every build. The frequency set could still drift during training, because the encoder in front of the circuit is trainable; it does not. The encoder drift $\|A - I\|_F$ at the end of training is 0.13 with one copy and 0.10 with eight, below the pre-registered 0.2 at which the design card would stop describing the trained model.

What this means. Proposition 1 and the ternary depth rule are not only theory: the design card computed before training is a correct, checkable description of the model after training. Every later statement about “inside” and “outside” Ω rests on this.

5.2 Designed frequencies beat random ones

The sharpest test of SMCD changes only one thing: the circuit, its size and its training stay fixed, and the encoding scalings are either SMCD’s or random draws from $U(0.5, 3)$. A worked example shows what is at stake. With four layers, random scalings of at most 3 reach frequencies of at most $4 \times 3 = 12$ radians per unit, while the heat and Poisson solutions contain a $15\pi \approx 47$ wave; SMCD’s scalings $\pi \cdot \{1, 3, 9, 27\}$ reach $40\pi \approx 126$. The random circuit cannot represent the fast wave at any parameter setting. The design wins on every seed of every comparison:

- **Heat, one copy**: $6.5\times$ lower error on all five seeds (Figure 2). The designed circuit is also the most consistent model in the study, its worst and best seeds within 4% of each other, while its random twin varies by $7.4\times$.

- **Poisson, eight copies:** $236\times$ (0.0037 against 0.87), all five held-out seeds.
- **Groundwater, eight copies:** $9.9\times$ (0.09% against 0.92%), all five held-out seeds (Section 6).

How robust is the heat result? Resampling the five paired seeds 5,000 times gives a 95% bootstrap interval of $2.5\times$ to $17.7\times$ for the error ratio, entirely above the pre-registered $1.5\times$ bar. Removing any one seed leaves the ratio above the bar all five times, so no single seed carries the result. With five pairs the smallest possible Wilcoxon p is 0.0625 however large the effect; tests without that floor make the strength visible: a paired t -test on log error gives $p = 0.006$ and a Mann–Whitney test $p = 0.008$.

What this means. This is the central claim of the method, and it holds on every problem where it was tested: choosing the encoding from the PDE’s symbol is worth one to two orders of magnitude in accuracy over the same circuit with an arbitrary encoding, at no cost in parameters.

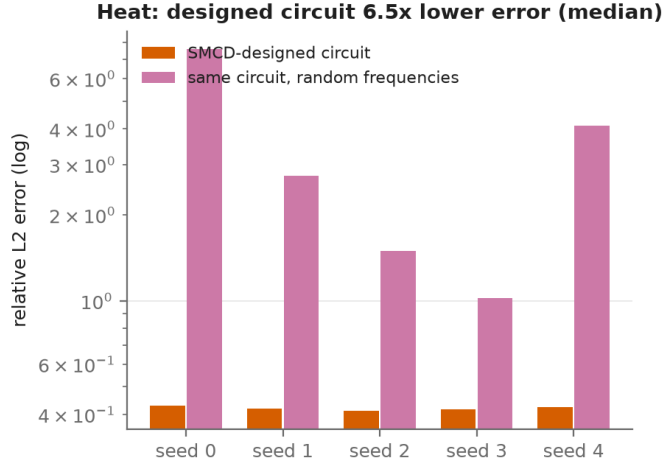


Figure 2: Heat equation, one copy: the circuit with SMCD’s scalings against the same circuit with random scalings, every seed.

5.3 The mechanism in the frequency domain

The improvement lands on the designed frequencies. Projecting the error onto Fourier modes at every checkpoint and accumulating how much better the circuit does than the MLP at each frequency, 99.9% of the circuit’s total improvement on Poisson lies inside Ω with one copy, 99.9% with eight, and 99.4% on groundwater; the pre-registered bar was 70%. Figure 3 shows it directly: the MLP’s error at high $|\omega|$ persists through training (the spectral-bias staircase), while the circuit clears the rows marked by Ω .

The NTK is flatter inside Ω . Proposition 4 says the quantum block’s kernel keeps non-decaying eigen-directions at the encoded frequencies, so its spectrum should be flatter inside Ω than outside, the opposite of a classical network’s power-law decay. With eight copies on Poisson the decay exponent inside Ω is flatter than outside by 15.2, against a pre-registered bar of 0.5.

What this means. Both instruments see the same thing from two sides: the circuit learns fast exactly where the design puts its frequencies. The accuracy gains come from the mechanism SMCD is built on, not from an incidental property such as size or initialisation.

SMCD also says when not to use a circuit. On the heat equation diffusion damps each Fourier mode at a rate proportional to ω^2 , so the 15π ripple in the initial condition vanishes almost immediately. SMCD’s design card, computed before any training, predicts a benefit of only 2.5% from a quantum layer,

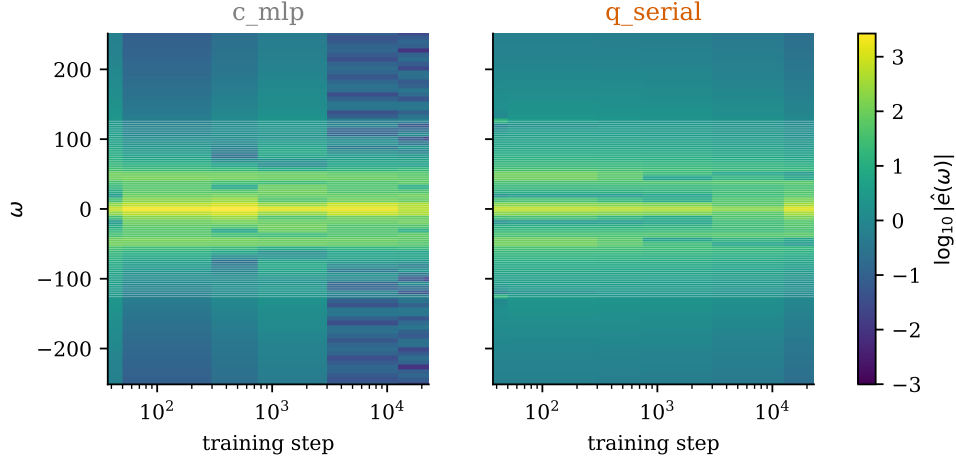


Figure 3: Poisson: per-frequency error $|\hat{e}(\omega)|$ over training, MLP (left, `c_mlp`) against the SMCD circuit (right, `q_serial`), seed 0, with Ω overlaid as thin white lines.

and the pre-registered check confirms it: no model improves on a plain MLP by more than 5%. A user learns this from the equation alone, before spending any compute.

No barren plateau at this scale. Gradient variance fitted against qubit count (4 and 6 qubits, three seeds) has slope 0.31, under the pre-registered $\log 2 = 0.69$ bar: no decay is detectable between 4 and 6 qubits. SMCD’s shallow depths, local observable and small initial angles are designed to avoid the plateau, and at this scale they do.

5.4 Circuit copies

What one circuit cannot do. The Poisson solution is $\sin \pi x + 0.3 \sin 15\pi x$. Tracking the error at those two frequencies (Figure 4), a single circuit cuts its 15π error to a median 10% of its starting value by step 1,000, faster than any classical model in the study (Fourier features 103%, MLP 86%). Its π error falls to about 30% by step 5,000 and then *rises* to about twice its starting value by step 20,000, on all five seeds.

A worked number explains the trade. The PDE residual contains u'' , and differentiating $\sin \omega x$ twice multiplies it by ω^2 , so an error at 15π costs $15^2 = 225$ times as much loss as the same error at π . All 81 Fourier amplitudes of a single circuit are functions of the same ten angles, read out through one expectation value, so moving the angles to fix the 15π amplitude also moves the π amplitude, and the optimiser accepts that because the loss it saves at 15π is 225 times larger. The loss stalls near 3.3×10^4 from step 1,000 while the error keeps moving, and all five seeds converge to the same trade-off.

Copies decouple the amplitudes. With K copies the head has K independent weights on K independently parametrised circuits, all with the same Ω (Section 3.8). On the tuning seeds the error falls with the number of copies (Figure 5); at the better of the two learning rates it is 1.67 with one copy, 1.27 with two, 0.08 with four and 0.003 with eight, almost three orders of magnitude.

Eight copies on held-out seeds. On seeds 10–14, which played no part in choosing the model, the eight-copy circuit reaches a median error of 0.0037, with every seed below 0.02. Five pre-registered claims hold on all five seeds: it beats the Fourier-feature network by three orders of magnitude and the same circuit with random frequencies by $236\times$; 99.9% of its improvement over the MLP lies inside Ω ; its NTK is flatter inside Ω by 15.2; and its encoder drifts by only 0.10. Every circuit seed trained stably at the selected learning rate.

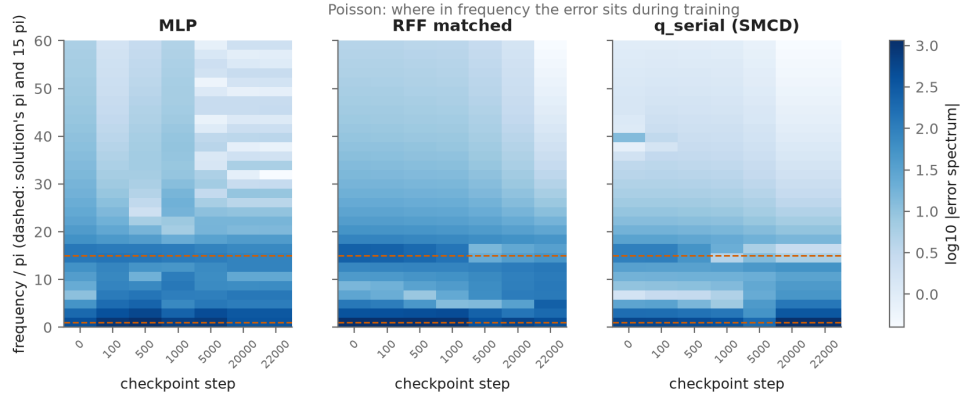


Figure 4: Poisson, one copy: error spectrum at each checkpoint (seed 0), with the error at π and 15π tracked over training.

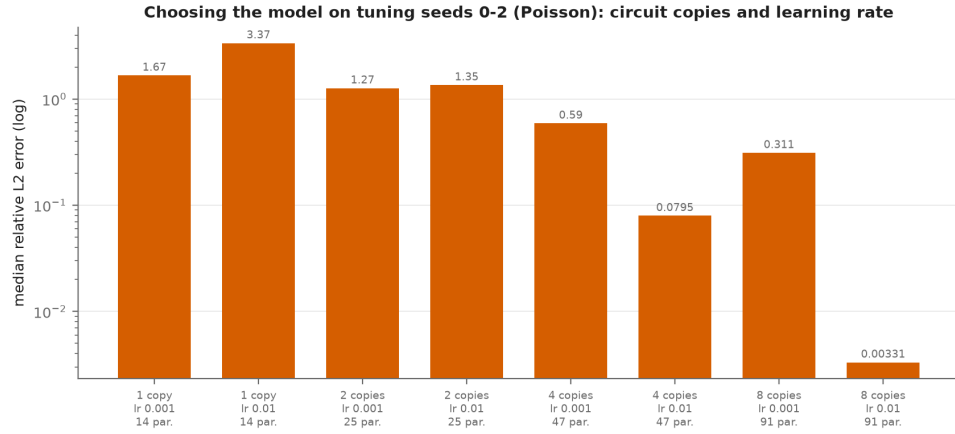


Figure 5: Model selection on Poisson, tuning seeds 0-2: median error for 1, 2, 4 and 8 copies at two learning rates.

6 Application: waterlogging under canal-irrigated farmland

SMCD needs a solution’s frequencies before training. Real problems where those are known in advance come from regular structure: evenly spaced sources, periodic geometry. This section applies the method to one such problem, with its claims pre-registered like all others (Section 4.3).

6.1 The problem

Two rivers $L = 2$ km apart bound a confined aquifer beneath irrigated farmland. Water enters from background recharge R_0 (rain and irrigation return flow) and from six canals running parallel to the rivers at equal spacing, each leaking q m³/day per metre of its length into a strip of width σ . Dupuit’s steady one-dimensional equation for the hydraulic head h (water-table height above the river datum) is

$$T h''(X) = -R(X), \quad R(X) = R_0 + \sum_{j=1}^6 \frac{q}{\sigma\sqrt{2\pi}} e^{-(X-c_j)^2/2\sigma^2}, \quad h(0) = h_A, \quad h(L) = h_B, \quad (1)$$

with $c_j = (j - \frac{1}{2})L/6$. The scenario uses values within textbook ranges [4, 13], not calibrated to one site: transmissivity $T = 100$ m²/day (silty sand), $R_0 = 0.4$ mm/day, $q = 0.4$ m²/day, $\sigma = 30$ m, $h_A = 1$ m, $h_B = 0$. On $x = X/L$ this is $-u'' = f$, the Poisson operator with non-zero boundary values, so every model and instrument applies unchanged. The canal spacing fixes the frequencies, harmonics of $2\pi \cdot 6$ in x , and SMCD designs a one-qubit, four-layer circuit for them.

Because $\sigma(z\Phi(z) + \phi(z))$ with $z = (X - c)/\sigma$ has second derivative $\phi(z)/\sigma$, the head has a closed form. An independent second-order finite-difference solver agrees with it to 5×10^{-8} m and converges to it at measured order 2.00 as the grid is refined from 100 to 6,400 intervals (Figure 7), which checks the closed form, the forcing and the boundary treatment together.

6.2 The practical question

A field is waterlogged where the water table rises within 1.5 m of the land surface, into the crop root zone. With the surface 9 m above the river datum, the water table peaks at 8.53 m and 713 m of the 2 km field is waterlogged (Figure 6). The solution answers the planning questions a farmer or irrigation engineer would ask (Figure 7):

- **Lining the canals.** Cutting canal seepage by 20% dries the whole field; a 10% cut reduces the waterlogged length from 713 m to 494 m.
- **Soil.** The same canals waterlog 1,778 m on clay loam ($T = 25$ m²/day) and none of the field on fine sand ($T \geq 150$).
- **Spacing.** Concentrating the same total seepage into two canals waterlogs 1,066 m and raises 84 cm mounds under each canal; spreading it over twelve reduces the mounds to 1 cm.

6.3 Can the trained model answer it?

The eight-copy SMCD circuit was trained on held-out seeds 10–14 at the full training budget, against a random-frequency circuit, an MLP and a frequency-matched classical model all of the same size, and the large MLP (Table 2, Figure 8). Five claims about it were pre-registered, and all five hold on every seed.

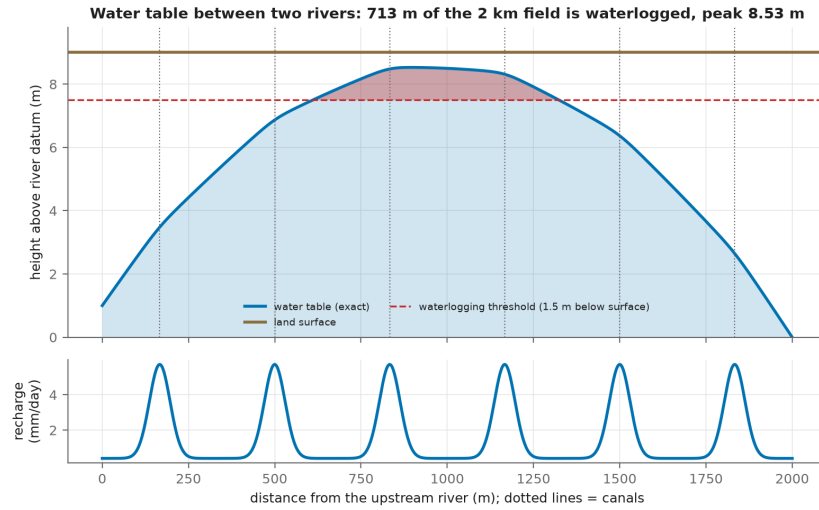


Figure 6: Exact water table between the two rivers (top) and the recharge that drives it (bottom). Red: the 713 m of field where the water table is within 1.5 m of the surface.

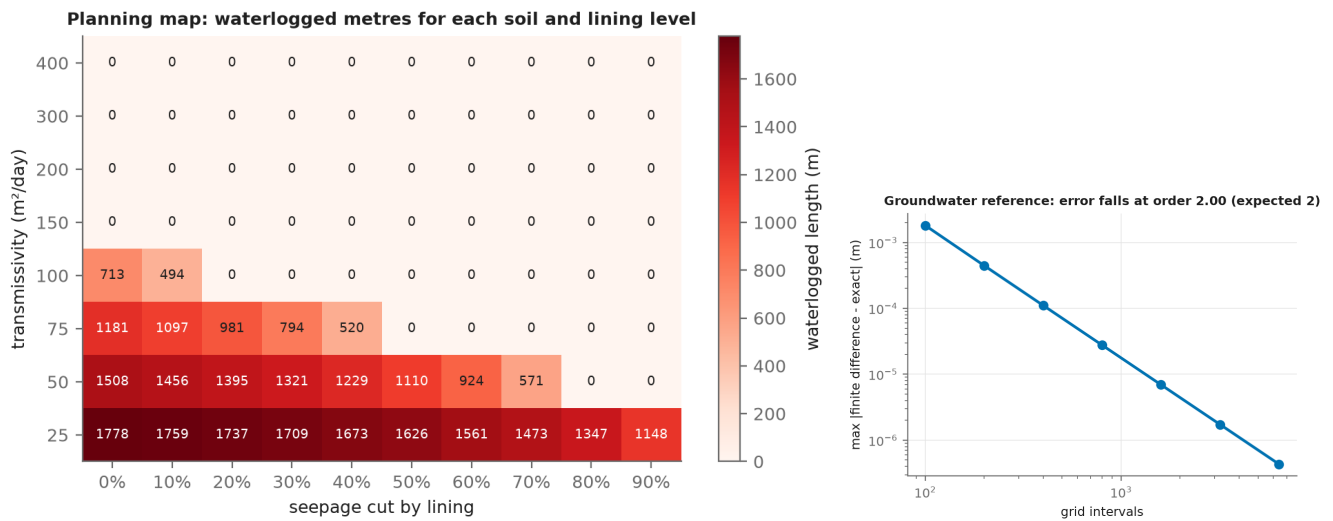


Figure 7: Left: waterlogged length for each combination of soil (transmissivity) and canal lining (seepage cut). Right: finite-difference error against the closed form as the grid refines.

Pre-registered claim, held on all five seeds			Measured
Error below 1%			0.09%
Beats a classical network of the same size ($\geq 1.3\times$)			18.2 $\times$
Beats the same circuit with random frequencies ($\geq 1.5\times$)			9.9 $\times$
Gain over the MLP lies inside Ω ($\geq 70\%$)			99.4%
Waterlogged length within 5% of the exact 713 m			714 m

Model	parameters	median error	waterlogged (m)
SMCD circuit, 8 copies	91	0.09%	714
same circuit, random frequencies	91	0.92%	738
MLP, same size	91	1.68%	672
frequency-matched classical, same size	83	22.93%	0
MLP, large	8,513	1.94%	707

Table 2: Groundwater, held-out seeds 10–14: the pre-registered claims for the SMCD circuit with eight copies (top) and every model’s median relative L^2 error and waterlogged length (bottom). Exact waterlogged length: 713 m.

Accurate to millimetres. The median error is 0.09%, with every seed below 0.5%. On a water table that peaks at 8.5 m, 0.09% is a typical error of a few millimetres, far below what any field measurement could resolve. The circuit is $18\times$ more accurate than an MLP with the same 91 parameters and $9.9\times$ more accurate than the same circuit with random frequencies, both on every paired seed, and it is also more accurate than the 8,513-parameter MLP, with about 1% of its parameters. 99.4% of its improvement over the MLP lies inside Ω , so the canal harmonics SMCD read off the forcing are where the accuracy comes from.

It answers the practical question. Relative error is not what a planner needs; the waterlogged length is. The circuit places it at 714 m against the exact 713 m. The same-size MLP, with 1.7% error, gives 672 m, 6% short, because a small error near the 7.5 m threshold moves where the water table crosses it. Getting the planning decision right requires accuracy well below 1%, and the circuit delivers it.

Why the circuit beats fixed features here. The classical model handed the same frequencies reaches 22.9% error and predicts no waterlogging. A plausible reason, not tested here, is that the solution is a broad mound plus small canal ripples: a linear head on fixed sine features at the canal harmonics represents the ripples but not the mound, while the circuit’s trainable encoder can shift and scale its frequencies to fit both.

7 Limitations

1. **Simulation only, small circuits.** Every circuit runs on a statevector simulator; nothing here ran on quantum hardware, and noise is modelled by analytic surrogates checked against a density-matrix simulation. Each circuit is one qubit per copy without entanglement, which a classical computer evaluates cheaply, and simulation cost capped the experiments at 6–8 qubits. No claim of quantum computational advantage is made or tested.
2. **Scope.** The evidence covers one-dimensional problems and one applied scenario, with values from textbook ranges rather than a calibrated site.
3. **Statistical power.** Five seeds give a smallest Wilcoxon p of 0.0625, so a confirmed comparison means all five seeds agree; bootstrap intervals and alternative tests support the central result but do not raise the sample size.

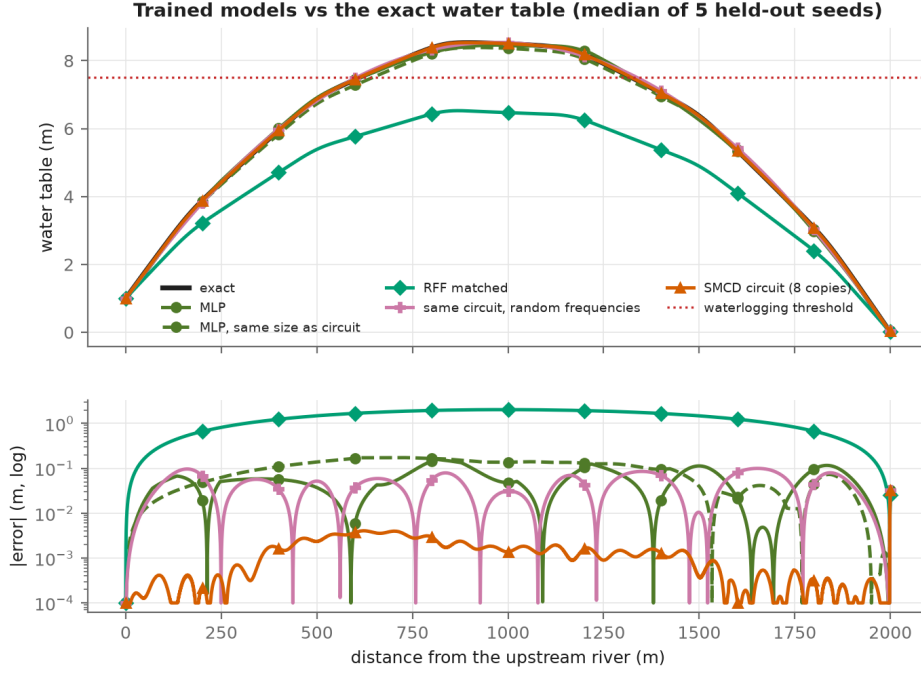


Figure 8: Predicted water table of each model against the exact solution (median of seeds 10–14), with the absolute error along the field below; the SMCD circuit stays within a few millimetres. Dotted: the waterlogging threshold.

4. **Tuning.** The eight-copy model was selected from a small grid of copy counts and learning rates, while classical families received a learning-rate search only.
5. **The affine encoder.** The encoder is restricted to an affine map so that Ω is well defined; a nonlinear encoder would make the frequency set input-dependent.

8 Conclusion

Spectrum-Matched Circuit Design turns the choice of a quantum circuit for a PDE into arithmetic: the operator’s Fourier symbol and the forcing give the frequencies the solution needs, and the encoding scalings, depth, qubit count and entangler follow by formula. The circuit it builds produces exactly the designed frequencies, and those frequencies matter: against the same circuit with random ones, the designed circuit is $6.5\times$ more accurate on the heat equation, $236\times$ on Poisson and $9.9\times$ on groundwater, on every seed.

The explainability instruments show how. The improvement lies on the designed frequencies, where the NTK is flat, and a single circuit’s Fourier amplitudes are coupled through shared angles, so it trades a slow wave for a fast one. Mixing eight copies of the designed circuit through a linear head keeps the frequency set and frees the amplitudes. On waterlogging of farmland from leaking canals, that model reaches 0.09% error, $18\times$ more accurate than a classical network of the same size, and answers the planning question to within 1 m. SMCD also tells a user in advance, from the equation alone, when a quantum layer will not help.

The most direct next steps are running the designed circuits on quantum hardware, extending SMCD to problems of higher dimension, and applying it to further problems whose structure fixes their frequencies in advance, as the canal spacing does here.

A Reproducibility

The code, every run’s metrics and instrument records, and the scripts that turn them into every figure, table and verdict are in the project repository; no figure reloads a trained model. All runs used one machine (NVIDIA RTX 4090 Laptop GPU, 32 CPU cores) with Python 3.12, PyTorch 2.11 (CUDA 12.8), PennyLane 0.45, NumPy 2.4 and SciPy 1.18, pinned by uv. A single integer seeds Python, NumPy and PyTorch with deterministic algorithms, so two runs at the same seed on the same device give bit-identical metrics. Seeds: six-problem matrix 0–4; coverage and barren-plateau sweeps 0–2; model selection for circuit copies 0–2; held-out evaluation of the eight-copy circuit and all groundwater models 10–14. `tasks.py test` runs the test suite (about 4 minutes on CPU), `tasks.py figures` redraws every figure from the committed runs (about 30 s), `tasks.py repro-quick` trains all problems and families at reduced size (about 9 minutes on CPU), and retraining all 327 recorded runs from scratch takes about 130 hours on the machine above.

References

- [1] Amira Abbas, David Sutter, Christa Zoufal, Aurelien Lucchi, Alessio Figalli, and Stefan Woerner. The power of quantum neural networks. *Nature Computational Science*, 1:403–409, 2021.
- [2] Ziv Chen, Gal G. Shaviner, Hemanth Chandravamsi, Shimon Pissnoy, Steven H. Frankel, and Uzi Pereg. Quantum physics-informed neural networks for Maxwell’s equations: circuit design, “black hole” barren plateaus mitigation, and GPU acceleration. *Quantum Machine Intelligence*, 8:21, 2026. doi: 10.1007/s42484-026-00365-w. arXiv:2506.23246.
- [3] Salah A. Faroughi and Farinaz Mostajeran. Neural tangent kernel analysis to probe convergence in physics-informed neural solvers: PIKANs vs. PINNs. arXiv:2506.07958, 2025.
- [4] R. Allan Freeze and John A. Cherry. *Groundwater*. Prentice-Hall, Englewood Cliffs, NJ, 1979.
- [5] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, 2018.
- [6] Said Lantigua, Gilson Giraldo, and Renato Portugal. A classical-quantum hybrid architecture for physics-informed neural networks. arXiv:2511.07216, 2025.
- [7] Jarrod R McClean, Sergio Boixo, Vadim N Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018.
- [8] M. Z. Naser. Fundamental flaws of physics-informed neural networks and explainability methods in engineering systems. *Computers & Industrial Engineering*, 212:111704, 2026. doi: 10.1016/j.cie.2025.111704.
- [9] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [10] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. The effect of data encoding on the expressive power of variational quantum machine learning models. *Physical Review A*, 103(3):032430, 2021.
- [11] Alexandr Sedykh, Maninadh Podapaka, Asel Sagingalieva, Karan Pinto, Markus Pflichtsch, and Alexey Melnikov. Hybrid quantum physics-informed neural networks for simulating computational fluid dynamics in complex shapes. *Machine Learning: Science and Technology*, 5(2):025045, 2024. doi: 10.1088/2632-2153/ad43b2.
- [12] Matthew Tancik et al. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, 2020.
- [13] David Keith Todd and Larry W. Mays. *Groundwater Hydrology*. John Wiley & Sons, Hoboken, NJ, 3 edition, 2005.
- [14] Kaveh Zabihi, Hamid Montazeri, and Akke S. J. Suiker. Hybrid quantum-classical physics-informed neural networks for solving nonlinear PDEs: when and where hybridization is effective? arXiv:2606.04679, 2026.